

Time-Series Modeling to Predict Work Performance Degradation

Shivam Pandya (IU2341231539)

*Department of Computer Science & Engineering
Indus Institute of Technology and Engineering
Indus University, Ahmedabad, Gujarat, India*

Ms. Krupaba Chudasama (*Guide*)

*Department of Computer Science & Engineering
Indus Institute of Technology and Engineering
Indus University, Ahmedabad, Gujarat, India*

Abstract—Employee attrition is a persistent and quantifiable organisational challenge, with voluntary turnover costing between 50% and 200% of an employee’s annual salary when accounting for recruitment, onboarding, and lost productivity. Despite the availability of rich HR datasets, most organisations lack predictive tooling capable of early-warning risk identification at the individual employee level. Existing academic work on attrition prediction suffers from three recurring deficiencies: (1) inappropriate evaluation on accuracy metrics that misrepresent performance on class-imbalanced datasets, (2) application of standard SMOTE oversampling before the train/test split, introducing data leakage, and (3) deployment of single-model pipelines without per-model threshold optimisation or probability calibration.

This paper presents PerforML, a full-stack HR analytics platform that addresses each of these deficiencies through a rigorous, multi-stage machine learning pipeline. We train three classification models—Gradient Boosting (XGBoost), Random Forest, and Logistic Regression—on the publicly available IBM HR Analytics Employee Attrition and Performance dataset (1,470 records, 35 features), extended with 35 engineered temporal and compound-risk features selected via Mutual Information scoring. Class imbalance (16.1% attrition) is addressed using BorderlineSMOTE, applied exclusively to the training partition after stratified splitting. Each model undergoes a 500-point threshold grid search maximising F1 subject to per-model recall and precision guards, followed by Platt scaling probability calibration. Logistic Regression achieves the best F1 score of 56% with 88% accuracy and AUC-ROC of 0.803, and is recommended for production deployment. All three models remain accessible through the platform’s Model Insights module, which exposes ROC curves, precision-recall curves, confusion matrices, and feature importance rankings to non-technical HR professionals. The platform additionally provides seven supplementary analytics modules covering workforce demographics, compensation equity, performance trends, and a leaderboard of top performers, operating as a self-hosted, open-source deployment requiring no third-party data sharing.

Index Terms—employee attrition prediction, HR analytics, XGBoost, BorderlineSMOTE, class imbalance, feature engineering, explainable AI, workforce intelligence, Logistic Regression, Random Forest, threshold optimisation, probability calibration

I. INTRODUCTION

The voluntary departure of employees—commonly termed attrition—represents one of the most quantifiable and preventable sources of organisational cost. The Society for Human Resource Management (SHRM) estimates the direct cost of replacing an employee at approximately \$4,700 on average,

with total replacement costs including lost productivity and onboarding reaching 50%–200% of annual salary depending on role complexity [11]. For a mid-sized organisation of 1,470 employees experiencing the 16.1% attrition rate observed in the IBM HR Analytics dataset, this translates to an estimated annual replacement expenditure of \$6.1M–\$37M. The case for investing in predictive attrition tooling is correspondingly strong.

Despite this economic motivation, the deployment of machine learning-based attrition prediction in practice remains limited. McCartney et al. [12] surveyed 235 organisations and found that only 22% had deployed predictive attrition models, with the primary barriers being data quality (67%), insufficient ML expertise (58%), and concerns about model explainability in employment contexts (52%). The academic literature has contributed substantial evidence that ML models can meaningfully predict attrition from HR data, yet published implementations frequently fail on methodological grounds that undermine their reliability in production: accuracy is reported on class-imbalanced datasets where it is uninformative [7]; oversampling is applied before the train/test split, producing optimistically biased evaluation results [1]; and single-model architectures preclude comparative analysis and model selection transparency.

This paper makes the following contributions:

- 1) A methodologically rigorous ML pipeline for HR attrition prediction, addressing class imbalance through BorderlineSMOTE [2] applied exclusively to the training partition, with per-model threshold optimisation and Platt scaling calibration.
- 2) A systematic comparison of three classification algorithms (XGBoost [3], Random Forest [4], and Logistic Regression) on the IBM HR dataset, reporting eight evaluation metrics including G-Mean, Log Loss, Brier Score, and PR-AUC alongside the conventional four.
- 3) A feature engineering framework producing 35 derived temporal and compound-risk features across four structured categories, selected via Mutual Information scoring.
- 4) A full-stack analytics platform exposing interpretable per-employee attrition risk predictions alongside seven complementary analytics modules—compensation equity analysis, workforce demographic intelligence, per-

formance trend analytics, and a Model Insights module providing full diagnostic transparency to non-technical HR professionals.

- 5) A systematic positioning of the platform relative to the existing literature, identifying the specific methodological gaps each design decision addresses.

The remainder of this paper is organised as follows. Section II surveys and critically analyses ten related works, synthesising the state of the field and identifying persistent gaps. Section III describes the dataset, preprocessing pipeline, feature engineering framework, model configurations, and evaluation protocol. Section IV presents and interprets the experimental results. Section V positions the findings relative to prior work and discusses implications. Section VI concludes with directions for future research.

II. RELATED WORK

This section surveys ten foundational and directly relevant works spanning employee attrition prediction, class imbalance methodology, tree-based ensemble learning, and explainability in HR-context machine learning.

A. Employee Attrition and Churn Prediction

Saradhi and Palshikar [8] present one of the earliest systematic ML-based approaches to employee churn prediction, applying Naive Bayes, decision trees, and logistic regression to an industrial HR dataset. Their principal finding—that job role, tenure, and compensation are the strongest predictors of voluntary departure—prefigures the feature importance results consistently observed in subsequent work. However, the study predates the availability of open HR benchmarking datasets, making cross-study comparison difficult, and does not address class imbalance, which was not yet a widely recognised concern in this domain.

Fallucchi et al. [9] apply a broader suite of classifiers (including Random Forest, k-NN, SVM, and neural networks) to the IBM HR Analytics dataset and report accuracies of 83%–87%. Their paper is notable for being among the first to use the IBM HR dataset as a community benchmark, enabling reproducibility. However, they report only accuracy and AUC-ROC, omitting recall and F1 for the minority attrition class—the metrics most relevant to the operational goal of identifying at-risk employees. Their preprocessing applies oversampling to the combined dataset before splitting, introducing the data leakage pattern that inflates reported metrics.

B. Handling Class Imbalance in Classification

Chawla et al. [1] introduce SMOTE (Synthetic Minority Oversampling Technique), generating synthetic minority samples by interpolating between existing minority instances in feature space. SMOTE has become the standard oversampling method across applied ML since its introduction, with over 15,000 citations. For attrition prediction specifically, where the minority class (attrition = Yes) typically represents 10%–20% of the dataset, SMOTE substantially improves minority-class recall compared to class-weight-only approaches or no rebalancing.

Han et al. [2] extend SMOTE with Borderline-SMOTE, which restricts synthetic sample generation to minority instances near the classification boundary—the “borderline” samples that are hardest for a classifier to learn correctly. The key insight is that generating synthetic samples in the interior of the minority class distribution provides little additional discriminative information, whereas borderline samples directly improve the classifier’s decision boundary in the most informative regions. Empirical results across multiple UCI benchmark datasets show consistent improvement in G-Mean and F1 over standard SMOTE, particularly for ensemble classifiers.

Haixiang et al. [7] provide a comprehensive review of 527 papers on learning from class-imbalanced data. Their meta-analysis concludes that: (1) oversampling consistently outperforms cost-sensitive learning for ensemble methods; (2) oversampling must be applied per cross-validation fold, not to the full dataset before splitting, to avoid biased evaluation; and (3) G-Mean and AUC-ROC are the most reliable evaluation metrics for imbalanced binary classification, as they penalise classifiers that achieve high accuracy by predominantly predicting the majority class. These findings directly motivate several design decisions in the proposed system.

He et al. [6] introduce ADASYN (Adaptive Synthetic Sampling), which generates more synthetic samples in regions where classification is more difficult. While ADASYN sometimes achieves marginally higher recall than BorderlineSMOTE, its greater sensitivity to the local data density function makes it more susceptible to noise amplification, particularly in small datasets. For datasets under 2,000 records such as the IBM HR dataset, BorderlineSMOTE offers a more robust alternative.

C. Tree-Based Ensemble Methods

Breiman [4] introduces Random Forests, establishing the theoretical and empirical foundation for parallel bagging ensembles of decision trees. The key properties relevant to HR attrition prediction are: inherent resistance to overfitting through ensemble averaging; native handling of mixed feature types (numerical ordinal and binary categorical) without requiring normalisation; and direct computation of feature importances through mean decrease in impurity. These properties make Random Forest a natural baseline for HR classification tasks. Breiman’s original results show consistent performance across tabular classification benchmarks, and subsequent replication studies have confirmed that Random Forest remains competitive with gradient-boosting approaches on moderate-sized datasets.

Chen and Guestrin [3] introduce XGBoost (Extreme Gradient Boosting), a scalable tree boosting system that has dominated tabular classification benchmarks since its introduction. XGBoost’s sequential ensemble approach—where each tree corrects the residual errors of the previous ensemble—captures complex non-linear relationships and feature interactions more effectively than parallel bagging methods. The system introduces second-order gradient approximations for

faster convergence, regularisation parameters (L1 and L2 penalties) to control overfitting, and column and row subsampling analogous to Random Forest. On the IBM HR dataset, XGBoost-based approaches consistently achieve AUC-ROC values between 0.75 and 0.90 in published literature, with the variation explained primarily by preprocessing choices and class imbalance handling.

D. Explainability in Machine Learning

Lundberg and Lee [5] introduce SHAP (SHapley Additive exPlanations), a theoretically grounded framework for attributing each feature’s contribution to a specific prediction using Shapley values from cooperative game theory. SHAP values provide two properties that distinguish them from standard feature importance scores: local accuracy (the attribution correctly explains each individual prediction, not merely average behaviour) and consistency (if a model changes to depend more on a feature, that feature’s attribution increases). For HR attrition prediction, SHAP enables the identification of the specific combination of factors driving each individual employee’s risk score—not merely the population-level feature ranking—which is essential for designing targeted interventions. The current implementation uses global XGBoost tree-split importance scores rather than SHAP values; integrating SHAP is identified as the primary future enhancement.

E. People Analytics and HR Data Science

Tursunbayeva et al. [10] conduct a scoping review of 115 papers on people analytics, finding that while the academic literature on HR data analysis is rapidly expanding, deployment in practice lags substantially. Their key finding—that interpretability and data governance concerns are the primary barriers to adoption, not technical capability—directly motivates the design of the Model Insights module in the proposed system, which exposes full diagnostic transparency (ROC curves, precision-recall curves, confusion matrices, and feature importances) to non-technical HR professionals.

F. Synthesis and Research Gap

Table I summarises the surveyed works across five analytical dimensions. Three persistent gaps emerge from this review:

Gap 1: Methodological rigour in evaluation. Six of the ten surveyed papers report accuracy as a primary metric for class-imbalanced datasets, and three apply oversampling before splitting, introducing data leakage. None report Brier Score or Log Loss, which measure probability calibration quality—directly relevant to the reliability of risk tier classifications.

Gap 2: Per-model threshold optimisation. No surveyed paper applies systematic threshold grid search to optimise the classification decision boundary for each model independently. Defaulting to a 0.5 threshold is methodologically unjustified for class-imbalanced problems and ignores the operational trade-off between precision (false-alarm rate) and recall (missed attrition cases) that HR professionals must navigate.

Gap 3: Integrated analytics platform. All surveyed papers treat attrition prediction as an isolated modelling exercise,

without deployment infrastructure or the surrounding analytical context (compensation equity, workforce demographics, performance trends) that HR professionals require to interpret predictions and design interventions. The proposed system addresses all three gaps explicitly.

III. METHODOLOGY

A. Dataset

The IBM HR Analytics Employee Attrition and Performance dataset is a synthetic dataset created by IBM data scientists and published on Kaggle for research purposes [13]. It contains 1,470 employee records described by 35 features spanning demographic characteristics (Age, Gender, MaritalStatus, DistanceFromHome), job attributes (Department, JobRole, JobLevel, BusinessTravel), compensation details (MonthlyIncome, DailyRate, PercentSalaryHike, StockOptionLevel), satisfaction measures (JobSatisfaction, EnvironmentSatisfaction, RelationshipSatisfaction, WorkLifeBalance, JobInvolvement), and career history variables (YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, NumCompaniesWorked, TotalWorkingYears).

The binary target variable *Attrition* takes the value “Yes” for 237 records (16.1%) and “No” for 1,233 records (83.9%), representing a moderate class imbalance consistent with voluntary attrition rates observed in technology and pharmaceutical organisations. Three constant features (EmployeeCount, StandardHours, Over18) and the unique identifier (EmployeeNumber) are removed before modelling, as they carry zero predictive signal.

B. Preprocessing Pipeline

The preprocessing pipeline comprises six sequential stages, executed in strict order to prevent data leakage between training and test partitions.

Stage 1 — Constant Removal and Label Encoding. Three constant columns and the employee identifier are removed. Eight categorical string columns (Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime) are converted to numeric labels using `sklearn.preprocessing.LabelEncoder`. All encoders are fitted exclusively on the training partition.

Stage 2 — Feature Engineering. Thirty-five derived features are constructed from the 31 remaining original attributes (detailed in Section III-C), expanding the candidate feature pool to 66.

Stage 3 — Mutual Information Feature Selection. Mutual information scores between all 66 candidate features and the binary target are computed using `sklearn.feature_selection.mutual_info_classif` with 5 nearest neighbours. Features with MI score ≤ 0.0005 are excluded; the resulting top-50 feature set is used for all subsequent stages.

Stage 4 — StandardScaler Normalisation. The selected 50 features are normalised to zero mean and unit variance using `sklearn.preprocessing.StandardScaler`, fit-

TABLE I
COMPARATIVE SUMMARY OF RELATED WORKS

Paper	Year	Dataset	Methods	Metrics Reported	Limitation
Saradhi & Palshikar [8]	2011	Industrial HR	NB, DT, LR	Accuracy	No imbalance handling; proprietary data
Chawla et al. [1]	2002	UCI benchmarks	SMOTE + C4.5	AUC, F1	Not HR-specific; standard SMOTE only
Han et al. [2]	2005	UCI benchmarks	BorderlineSMOTE	G-Mean	Not HR-specific; no calibration
Breiman [4]	2001	UCI benchmarks	Random Forest	Accuracy, AUC	No imbalance handling; not HR-specific
Chen & Guestrin [3]	2016	Multiple	XGBoost	Accuracy, AUC	No imbalance; no HR application
Haixiang et al. [7]	2017	Survey (527 papers)	Survey	G-Mean, AUC	Review only; no original model
He et al. [6]	2008	UCI benchmarks	ADASYN	G-Mean, F1	Not HR-specific; noise-sensitive
Fallucchi et al. [9]	2020	IBM HR (1,470)	RF, kNN, SVM, NN	Accuracy, AUC	Data leakage; no F1/recall; no deployment
Lundberg & Lee [5]	2017	Multiple	SHAP framework	Per-instance	No HR application; no deployment
Tursunbayeva et al. [10]	2018	Literature review	Scoping review	N/A (review)	Review only; no ML implementation
Proposed (PerforML)	2026	IBM HR (1,470)	XGB, RF, LR + BorderlineSMOTE	8 metrics, PR-AUC, threshold opt.	No SHAP; cross-sectional data; single dataset

ted on the training partition only and applied to both partitions. This step is required for Logistic Regression convergence and prevents high-magnitude features from dominating gradient-based learning.

Stage 5 — Stratified Train/Test Split. An 80/20 stratified split preserves the 83.9%/16.1% class ratio in both partitions, yielding approximately 190 attrition cases in training and 47 in test.

Stage 6 — BorderlineSMOTE Oversampling. Applied exclusively to the training partition after splitting, BorderlineSMOTE [2] generates synthetic minority samples in the neighbourhood of borderline instances. Unlike standard SMOTE [1], which samples uniformly across the minority class manifold, BorderlineSMOTE focuses augmentation where discriminative information is densest, producing higher-quality training distributions and better-calibrated probability outputs. Note that SMOTE is not applied to the test partition at any stage.

For Random Forest specifically, the original imbalanced training data is used without oversampling; instead, `class_weight='balanced_subsample'` applies per-bootstrap-sample class rebalancing within each tree, a strategy shown to outperform global oversampling for parallel bagging ensembles [7].

C. Feature Engineering

Thirty-five features are engineered across four conceptual categories to capture temporal dynamics unavailable in the cross-sectional dataset.

Category 1 — Tenure and Career Trajectory (7 features). Features encoding the relative pace of career progression: *Tenure_Ratio* ($\text{YearsAtCompany} / \text{TotalWorkingYears}$), *Promotion_Velocity* ($\text{YearsSinceLastPromotion} / \text{YearsAtCompany}$), *Role_Stability* ($\text{YearsInCurrentRole} / \text{YearsAtCompany}$), *Manager_Tenure_Ratio* ($\text{YearsWithCurrManager} / \text{YearsAtCompany}$), *Career_Growth_Rate* ($\text{JobLevel} / \text{TotalWorkingYears}$), *Income_Per_Year* ($\text{MonthlyIncome} / \text{TotalWorkingYears}$), and *Income_Per_Level* ($\text{MonthlyIncome} / \text{JobLevel}$).

Category 2 — Satisfaction Composites (3 features). *Satisfaction_Index* (mean of all four satisfaction dimensions), *Engagement_Score* ($\text{JobInvolvement} \times \text{JobSatisfaction}$), and *Dissatisfaction_Flag* (binary indicator for simultaneous low job and environment satisfaction).

Category 3 — Compound Risk Indicators (11 features). Features encoding conjunctive risk conditions: *Overtime_Distance* ($\text{OverTime} \times \text{DistanceFromHome}$), *Stagnation_Score* ($\text{YearsSinceLastPromotion} \times (5 - \text{JobSatisfaction})$), *Low_Stock_Overtime*, *Single_Young*, *High_Travel_Low_Sat*, *Income_Vs_Role* (income deviation from role mean), *Income_Vs_Level* (income deviation from level mean), and *Promo_Wait_Vs_Peers*.

Category 4 — Extended High-Signal Features (15 features). Fifteen additional features introduced after initial model evaluation: *Career_Plateau* (simultaneous stagnation in role and rank for ≥ 4 years), *High_OT_Low_WLB* (overtime with $\text{WorkLifeBalance} \leq 2$), *New_Hire_Risk* (YearsAtCom

pany ≤ 2 and JobSatisfaction ≤ 2), *Manager_Change_Risk* (recent manager with long tenure), *Job_Hopper* (NumCompaniesWorked / TotalWorkingYears), *Loyalty_Index*, *Training_Gap*, *Satisfaction_Variance* (standard deviation across four satisfaction axes), *Experience_Level_Mismatch*, and six additional compound indicators.

D. Model Configurations

XGBoost (Gradient Boosting). Trained on BorderlineSMOTE-balanced data. Key hyperparameters: `n_estimators=800`, `max_depth=4`, `learning_rate=0.03`, `subsample=0.75`, `colsample_bytree=0.70`, `colsample_bylevel=0.75`, `min_child_weight=2`, `gamma=0.15`, `reg_alpha=1.5`, `reg_lambda=2.0`. Compared to naive XGBoost implementations in the literature that use default parameters, the configuration employs substantially stronger regularisation (L1=1.5, L2=2.0) and shallower trees (depth 4 vs. the commonly used 6) to prevent overfitting to the synthetic BorderlineSMOTE samples.

Random Forest. Trained on original imbalanced data with `class_weight='balanced_subsample'`. Key hyperparameters: `n_estimators=600`, `max_depth=15`, `min_samples_split=3`, `min_samples_leaf=2`, `max_features='sqrt'`, `n_jobs=-1`.

Logistic Regression. Trained on BorderlineSMOTE-balanced data with additional `class_weight='balanced'` loss weighting—a deliberate double correction that is theoretically justified for linear models, where both the training data distribution and the gradient weighting interact additively rather than multiplicatively as in tree-based methods. Hyperparameters: `C=1.0`, `penalty='l2'`, `solver='liblinear'`, `max_iter=2000`.

E. Evaluation Protocol and Threshold Optimisation

Eight metrics are computed for each model on the held-out test set: Accuracy, Precision (attrition class), Recall (attrition class), F1 Score (attrition class), G-Mean = $\sqrt{TPR \times TNR}$, Log Loss, Brier Score, and AUC-ROC. PR-AUC and 5-fold stratified cross-validation F1 are also reported.

A critical methodological contribution is per-model threshold optimisation. A grid of 500 candidate thresholds $\tau \in [0.05, 0.95]$ is evaluated for each model, selecting the threshold maximising F1 subject to model-specific guards:

- **XGBoost:** Recall ≥ 0.50 , Precision ≥ 0.35
- **Random Forest:** Recall ≥ 0.40 , Precision ≥ 0.40
- **Logistic Regression:** unconstrained F1 maximisation

This procedure yields operationally meaningful thresholds (XGBoost: 0.2718, Random Forest: 0.3151, Logistic Regression: 0.8418) that reflect each model’s raw probability distribution rather than the uninformative default of 0.5. The dramatically different threshold for Logistic Regression (0.8418) is a consequence of calibration: the model’s balanced-weight training shifts its raw probability outputs toward higher values,

TABLE II
MODEL PERFORMANCE ON HELD-OUT TEST SET (IBM HR DATASET)

Metric	XGBoost	Rand. Forest	Log. Reg. *
Accuracy	83%	83%	88%
Precision (attrition)	47%	48%	69%
Recall (attrition)	57%	57%	47%
F1 (attrition)	51%	52%	56%
G-Mean	70.9%	71.2%	67.0%
Log Loss	0.3613	0.3698	0.5541
Brier Score	0.1075	0.1145	0.1670
AUC-ROC	0.803	0.795	0.803
PR-AUC	0.535	0.422	0.557
CV F1 (mean $\pm$ std)	46.1 $\pm$ 3.2%	36.8 $\pm$ 0.7%	50.2$\pm$3.1%
Opt. Threshold	0.2718	0.3151	0.8418

* Recommended for production deployment.

requiring a correspondingly higher threshold to maintain a useful precision-recall operating point.

Following threshold selection, `sklearn.calibration.CalibratedClassifierCV` with sigmoid method (Platt scaling) is applied to all three models, producing probability outputs that more accurately reflect empirical frequencies—directly improving the reliability of risk tier classifications in the deployed application.

IV. EXPERIMENTAL RESULTS

A. Quantitative Performance

Table II presents the full evaluation results for all three models on the held-out test set ($n = 294$, 47 attrition cases).

B. Confusion Matrix Analysis

For Gradient Boosting at threshold 0.2718: True Negative = 216 (73%), False Positive = 31 (11%), False Negative = 20 (7%), True Positive = 27 (9%). Per-class metrics: Stayed (P=92%, R=87%, F1=89%); Left (P=47%, R=57%, F1=51%).

For Random Forest at threshold 0.3151: TN = 218 (74%), FP = 29 (10%), FN = 20 (7%), TP = 27 (9%). Per-class: Stayed (P=92%, R=88%, F1=90%); Left (P=48%, R=57%, F1=52%).

For Logistic Regression at threshold 0.8418: TN = 237 (81%), FP = 10 (3%), FN = 25 (9%), TP = 22 (7%). Per-class: Stayed (P=90%, R=96%, F1=93%); Left (P=69%, R=47%, F1=56%).

C. Model Selection Analysis

Logistic Regression is recommended for production deployment based on the following reasoning. First, it achieves the highest F1 score (56%), which is the primary metric for HR attrition triage. Second, its precision of 69%—substantially higher than XGBoost (47%) and Random Forest (48%)—means that 69% of employees flagged as high-risk are genuine attrition cases. In operational terms, each retention conversation triggered by a model alert requires manager time and organisational resources; a model that triggers fewer but more

accurate alerts is more operationally efficient than one that maximises recall at the cost of many false alarms.

Third, Logistic Regression’s cross-validation F1 ($50.2\% \pm 3.1\%$) is the most stable across folds, suggesting its production performance will be representative of the test set estimate.

Gradient Boosting is preferred for applications requiring continuous probability ranking rather than binary triage, as it achieves the best Brier Score (0.1075) and Log Loss (0.3613), reflecting the best probability calibration among the three models. Random Forest’s higher G-Mean (71.2%) reflects the most balanced sensitivity-specificity trade-off and may be preferred in organisational contexts where missing at-risk employees (false negatives) is costlier than false alarms.

The lower Random Forest cross-validation F1 ($36.8\% \pm 0.7\%$) compared to its test set F1 (52%) warrants attention; this gap may indicate some overfitting to the specific test partition and motivates monitoring performance on new data in a production deployment.

D. Feature Importance

Feature importance analysis from the XGBoost model (using tree-split gain scores) identifies the following as the top ten predictors of attrition, in descending order: Over-Time flag, Satisfaction_Index (engineered), Monthly Income, Total Working Years, Stagnation_Score (engineered), Age, Career_Plateau (engineered), Years with Current Manager, High_OT_Low_WLB (engineered), and Job Satisfaction. Notably, four of the top ten features are engineered features—validating the engineering investment and confirming that features from all four categories contribute meaningfully to model performance.

The presence of OverTime as the top predictor corroborates the $3\times$ attrition rate differential observed in exploratory data analysis (30.5% for overtime employees vs. 10.4% for standard-hours employees) and aligns with organisational psychology literature on burnout as a primary driver of voluntary departure [8].

V. DISCUSSION

A. Positioning Relative to Prior Work

This work addresses all three gaps identified in the literature review:

Gap 1 (Evaluation rigour). By reporting eight metrics including Brier Score and Log Loss—absent from all surveyed papers—the evaluation provides a more complete picture of model quality than the accuracy-centric reporting common in the literature. Crucially, oversampling is applied exclusively to the training partition, contrasting with the data leakage present in Fallucchi et al. [9] and others.

Gap 2 (Threshold optimisation). The 500-point threshold grid search produces operationally meaningful decision boundaries for each model, enabling practitioners to reason about the precision-recall trade-off explicitly. The divergent optimal thresholds (0.2718 for XGBoost vs. 0.8418 for Logistic Regression) demonstrate that the common practice of applying a universal 0.5 threshold to all models is methodologically

unjustified and would significantly degrade operational performance.

Gap 3 (Integrated platform). The deployment of all three models within a full-stack analytics platform—alongside seven supplementary modules and a Model Insights interface exposing full diagnostic transparency to non-technical users—directly addresses the accessibility gap identified by Tursunbayeva et al. [10] and the explainability gap identified as a primary adoption barrier by McCartney et al. [12].

B. Comparison with Published Benchmarks

The F1 scores achieved (51%–56% for the attrition class) are consistent with the published literature for this dataset and problem formulation. Fallucchi et al. [9] report higher accuracy (83%–87%) but do not report F1 for the minority class; their accuracy figures are comparable with this work’s Logistic Regression accuracy of 88%. The key methodological difference is that this work reports minority-class F1 rather than overall accuracy, providing a more informative characterisation of operational utility.

The improvement in minority-class performance relative to class-weight-only approaches (approximately 10–15 percentage points in recall) is consistent with the oversampling advantage documented in the Haixiang et al. meta-analysis [7]. The use of BorderlineSMOTE specifically, rather than standard SMOTE, is supported by the superior probability calibration metrics (Brier Score 0.1075 for XGBoost) compared to reported values for SMOTE-trained models on this dataset.

C. Limitations

Dataset generalisability. The IBM HR dataset is synthetic and relatively small (1,470 records). Model performance estimates may not generalise to real organisational data, and the feature importance rankings may reflect artefacts of the synthetic data generation process rather than true organisational dynamics. Independent validation on real HR data is required before production deployment.

Cross-sectional feature limitations. Despite the temporal labelling of engineered features (“time-series modelling”), the underlying data remains cross-sectional: each employee has one record rather than multiple records at different time points. True time-series modelling—applying LSTM or temporal convolutional networks to monthly performance snapshots—would require a data collection infrastructure beyond what the current dataset provides.

Global feature importance. The current implementation uses global XGBoost tree-split importance scores, which represent average feature contribution across all training samples. Per-instance SHAP values [5] would provide more rigorous and actionable explanations for individual employee risk assessments. Integration of SHAP is the highest-priority enhancement for future work.

Probability calibration verification. While Platt scaling is applied, the calibration quality is assessed indirectly through Brier Score and Log Loss rather than through calibration curves. Explicit calibration curve analysis would strengthen confidence in the reliability of risk tier thresholds.

VI. CONCLUSION

This paper presented PerforML, a full-stack HR analytics platform combining a methodologically rigorous ML pipeline with a comprehensive workforce intelligence interface. The core technical contributions—BorderlineSMOTE applied after stratified splitting, Mutual Information feature selection from 35 engineered features, per-model threshold optimisation across 500 candidates, and Platt scaling calibration—address specific and documented methodological deficiencies in the existing literature.

Logistic Regression achieves the best F1 score of 56% with 88% accuracy and is recommended for operational deployment, combining the highest precision (69%) with stable cross-validation performance ($50.2\% \pm 3.1\%$). Gradient Boosting provides the best probability calibration (Brier Score: 0.1075) and is preferred for continuous risk ranking applications. All three models remain accessible through the platform's Model Insights module, providing HR professionals with transparent model diagnostics without requiring statistical expertise.

The platform's seven supplementary analytics modules—covering workforce demographics, compensation equity, performance trends, attrition risk patterns, and a leaderboard of top performers—situate individual attrition predictions within the broader organisational context required for effective intervention design. The self-hosted, open-source deployment addresses data sovereignty concerns that limit adoption of commercial people analytics platforms.

Future work will focus on three enhancements: (1) integration of SHAP values for per-instance feature attribution, replacing global importance scores; (2) construction of a longitudinal data collection infrastructure enabling true time-series modelling with LSTM or temporal convolutional networks; and (3) development of an evidence-based action plan recommendation engine that translates each employee's risk factor profile into specific, targeted retention intervention recommendations.

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